PRE-REGISTRATION — CUDA Graphs / torch.compile null test

Registered: 2026-08-14. **No null-test data exists as of this writing.** Drafted by the planning assistant from decisions made in-thread; predictions and decision rules were established before drafting. Commitment is the author's:
COMMITTED BY: _____ (Muhit — read fully, edit predictions if yours differ, sign before the first null-test run. Unsigned = not registered.)

The question

Can generic launch tooling (CUDA Graphs, torch.compile) delete the measured ~80% GPU-idle of the MelBandRoformer front-end — or does it leave structure on the table that only a custom grouped ragged-K kernel can claim?

Environment (from 2026-08-14 probe session)

Rented RTX 5090 (SM120), RunPod PyTorch template: torch 2.8.0+cu128, triton 3.4.0, arch list includes sm_120.
HW performance counters LOCKED on this host type (ERR_NVGPUCTRPERM) — timing and nsys timelines unaffected; ncu sessions need a different host.
nsys ABSENT from template — setup.sh must install nsight-systems-cli before arms run.
`F.grouped_mm` absent at torch 2.8; `torch._grouped_mm` present but arch-gated off capability (12,0) — total exclusion, both raggedness patterns, all dtypes (probe, 2026-08-14).

Arms (all on ONE instance, ONE session; never mixed with office-box numbers)

arm	config	role
A	eager, stock `F.normalize` chain	baseline (regenerated on this host)
B	A + manual whole-forward CUDA Graphs capture	dispatch-deletion isolate
C1	`torch.compile(mode="reduce-overhead")`	reviewer's default answer
C2	`torch.compile(mode="max-autotune")`	Inductor's best fusion
D	hand-fused RMSNorm kernel, eager elsewhere	validates Inductor-parity claim
E	D + manual capture	fused + replay
F	A + multi-stream band loop	overlap without capture
G	**D + multi-stream + capture**	**honest counterfactual: best achievable without solving the representational gap**

Two regimes per arm: 8 s chunk (352,800 samples) and 100 ms streaming chunk (4,410).

Decision rule (pre-committed)

Computed at the 100 ms regime, on arm G: R = median wall-time / GPU-busy-time (busy from nsys kernel sum, same run).

R ≤ ~1.25 → gaps closed by overlap+replay; launch-collapse headroom is small. Candidate (a)'s quantitative case resizes to wave-quantization + reuse + the deleted stack copy; the categorical (representational-gap) story carries the claim.

R ≥ ~1.5 → measured headroom stands; grouped kernel has a quantitative target.

Between → gray zone; the week-4 gate weighs it against candidate-(b) evidence.

Structural criterion, independent of timing: if the best generic arm's trace still shows the ~60 separate GEMM kernels per pass, the categorical claim is confirmed by inspection regardless of wall-time.

Predictions (named, with mechanisms — falsifiable)

P1 — winning arm at 100 ms: G. Mechanism: fused norm removes ~300 launches' work; streams overlap the 60 GEMMs; capture deletes host dispatch of what remains.

P2 — C2 (max-autotune) fuses the norm chain (reduce→clamp→div→mul→mul → 1–2 kernels), reproducing the hand fusion; kernel count drops ~388 → ~120/pass. Mechanism: Inductor vertical fusion covers elementwise+reduction chains.

P3 — the 60 ragged-K Linears survive ALL compile modes as separate kernels. Mechanism: no Inductor IR vocabulary for runtime reduction-extent tables; group_batch_fusion requires identical shapes; `_grouped_mm`

arch-gated anyway. (Stack copy under compile: 0, 1, or 60 kernels — genuinely uncertain, flagged.)
P4 — complex-STFT breaks capture/compile. Dynamo graph-breaks at complex ops / stft; arm B whole-forward capture may fail ⇒ pre-authorized fallback: scope B/E/G capture to band_split stage only, stated in the log.
P5 — cross-host replication split. Structural facts replicate on the rented host: ~388 launches/pass, same kernel identities, GPU-busy ≈ 1.26 ms ± clock/vCPU variation. Host-dependent facts do not: wall time and idle fraction move; predicted direction — cloud vCPU dispatches slower than office CPU ⇒ idle fraction holds or RISES above 77–83%.
P6 — (eval-week, registered now) L2 sweep: the traffic-deletion share of the grouped kernel's win grows as L2 shrinks (A5000 6 MB → Hopper ~50 MB → 5090 96 MB); the launch-collapse share is L2-invariant.

Methodology constraints (violating any invalidates the run)

All arms, same instance, same session. Separate OS process per chunk regime.
`torch.compile(..., dynamic=False)`; let autotune fully settle before timing.
`TORCH_LOGS=perf_hints` (and recompile logs) captured on every compile arm.
Graphs mechanics: static I/O buffers, `.copy_()` into them, warmup on side stream, capture OUTSIDE the timed region.
≥ 50 timed iters after warmup; per-iteration times printed; median + min/max; ≥ 3 process restarts for the headline numbers. **5a. TIMING PROTOCOL (clarification registered 2026-08-16, pre-data):** "wall" for the decision rule = per-iteration latency with `torch.cuda.synchronize()` after each pass. Rationale: streaming semantics — chunks arrive on real-time cadence, back-to-back pipelining is unavailable in deployment, so latency (not throughput) is the decision quantity in the 4410 regime. Each arm×regime ALSO records one free-running block (sync, N iters, sync; total/N) as `throughput_ms_per_pass` — the honest metric for the offline 8 s regime, the continuity bridge to the historical free-running baseline (5.5–7.5 ms), and the empirical measure of the serialization delta. Known bias: per-iteration sync can only inflate G's wall → inflates R → biases AGAINST the kill verdict (toward candidate (a) surviving); expected < 2%. Escalation clause: if the free-running delta at the 4410 regime exceeds 5%, the decision rule is evaluated under BOTH protocols and any disagreement between their verdicts is escalated, not silently resolved.
≥ 2-pass different-data validation on every arm — including D/E/G (stale-buffer aliasing under capture passes single-pass checks while being wrong).
First action of the session, before any timing: `torch.compile` **output_code reads at both chunk sizes** — kernel inventory recorded.
nsys timeline of (best generic arm) and (G): kernel count + gap structure. These two traces are the structural evidence either way.
P8 — full-model fraction (registered 2026-08-15, unmeasured on clean provenance). Session opener of null-test day, before any arms: one nsys trace of the FULL forward, 8 s chunk. Prediction: front-end fraction on the 5090 comes out ABOVE the ~5% work-data figure from the A5000 production host — ~8–15%. Mechanism: the numerator (front-end) is host-dispatch-bound and does not scale with the card; the denominator (bandwidth-bound transformer stack) shrinks with 1792 vs 768 GB/s. Also recorded from this trace: kernel-count and gap structure of the attention stack (candidate (b)'s first evidence). Gate-memo obligation attached: pin the quality-viable streaming operating point (likely 0.5–2 s hops, not 100 ms); the launch-cost argument is monotonic in chunk size and survives wherever the pin lands.

Version matrix (addendum registered 2026-08-15, before any newer-torch check has run)

cell	when	answers
SM120 × torch 2.8	DONE 2026-08-14	total arch exclusion, both patterns, all dtypes
SM120 × latest stable	venv after null-test arms, or micro-pod	gate evolution on consumer Blackwell
SM90 × latest stable	the planned Hopper session	gate on native arch + the deferred ragged-K expressibility question
(writing time)	once, before submission	claim-freshness sweep on then-current stable — no re-checks between these three

P7 — the SM120 arch gate PERSISTS in latest stable (F.grouped_mm present, device still rejected, all dtypes, both patterns). Mechanism: the op's kernels are Hopper wgmma/TMA implementations; no mma.sync consumer path exists — the selective- abandonment thesis as a forecast. Falsification consequences: gate lifted → in-tree op joins the SM120 baseline table, claim sentence re-scopes; gate lifted AND 2D×2D-offs-on-K runs on SM120 → major-revision flag, escalate immediately.
Environment rule: torch is NEVER upgraded on the null-test pod before or among the arms; version-matrix checks run in a post-arms venv or a separate micro-session.

What this test cannot decide

Whether the grouped kernel BEATS arm G — the kernel doesn't exist yet. This test bounds the headroom and confirms/kills the categorical claim; the quantitative comparison happens in eval against the same arm-G configuration.